

SAKSHAM JAIN

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EDUCATION

B.Tech.	2027	Delhi Technological University, New Delhi	8.7
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WORK EXPERIENCE

AI Research Intern | Massachusetts Institute of Technology (MIT CSAIL)

Jun 2025 - Dec 2025

Advisor: Prof. Manolis | *Remote*

- Stress-tested LLM-based Knowledge Graph pipeline using **100-parallel** semaphore async calls, extracting **3,000+ subject-relation-object triplets** across scientific textual datasets with **90%+ extraction accuracy**.
- Generated **node & edge embeddings** using Node2Vec (3 variants), RotatE, and GraphSAGE on **~2,500 processed entities** to benchmark clustering coherence and perform high-fidelity link prediction.
- Pruned **500+ hallucinated triples** via canonical entity aggregation, boosting KG precision and improving curation throughput by **25%**; enhanced React frontend with UMAP clustering, and cluster density control.

AI Software Intern | AmberFlux EdgeAI Pvt. Ltd.

Nov 2024 - Jan 2025

- Engineered a Retrieval-Augmented Generation (RAG) system on Azure, fine-tuning LLaMA with LoRA, and using LlamaIndex, LangChain, and vector databases for customer feedback analysis; **reduced model deployment time by 15%**.
- Achieved a classification score of 0.82 using a custom metric combining F1-score and distance-based evaluation, leveraging few-shot prompting for improved contextual accuracy.

PUBLICATIONS

- IllumiCurveNet**: Low-Light Image Enhancement of Lunar Permanently Shadowed Regions Using a Self-Guided Loss Framework | **International Joint Conference on Neural Networks (IJCNN) 2025 | First Author**
- Not All Pixels Sink: Phase-Guided Representation Learning For Underwater Image Restoration | **CVPR'26 Workshop**
- DECA-DiaXEL: An Explainable Ensemble Learning Framework for Early Diabetes Detection in ICU Settings Using a Dimensional Expansion-Contraction Architecture | International Conference on Computing, Communication and Learning (CoCoLe 2025)
- SleepWalk: A Three-Tier Benchmark for Stress-Testing Instruction-Guided Vision-Language Navigation | Under Review at COLM
- SARCIO: Competitive Patch-Level Safety Alignment for Text-to-Image Diffusion Models | Under Review at EMNLP

ACADEMIC PROJECTS

Real-Time Bidding Optimisation Engine for Adobe | Pytorch, NumPy, Machine Learning, Data Analysis.

- Processed RTB bid logs and crafted an ensemble model with Outer Product-based Neural Network and XGBoost, achieving AUC scores of 0.9094 (CTR) and 0.8889 (CVR) by making 2 different approaches based on advertiser_weight.
- Engineered a bidding algorithm using CTR/CVR predictions and value-adjusted formulas, executing under **3ms** and **200MB** to maximize the target score (clicks + N * conversions).

Unsupervised Image Enhancement (IJCNN'25) | Deep Learning, Pytorch, Image Enhancement.

- Architected IllumiCurveNet**, an unsupervised deep learning model for enhancing extreme low-light images of lunar PSR, integrating dilated convolutions, spatial attention, adaptive gamma correction with an Encoder-Decoder network.
- Created a **Self-Guided Loss Framework** with 2 novel loss functions, boosting texture and contrast in PSR images, outperforming existing methods in **no-reference image quality metrics** (PIQE, NIQE, BRISQUE) by **~29%**.

LyriSense: Sentiment-Driven Song Recommender | Sentiment Analysis, Cosine Similarity, Vector DB.

- Built a **song recommendation system** - Performed sentiment analysis, topic modeling, and generated embeddings on lyrics for 25k songs using LaBSE, DistilBERT-base and Facebook/hubert-large-ls960 models from Hugging Face.
- Implemented a custom weighted similarity function, optimized through hyperparameter tuning and achieved an accuracy of **95%** with a threshold of **0.8 similarity score** for a dataset of **3k users**.

ACHIEVEMENTS

- Winner at Adobe Devcraft** for creating a Real Time Bidding Optimisation Engine.
- Winner at Microsoft Code Crunch ML Challenge** for building a real time Facial Expression Recognition System.
- National Finalist at Smart India Hackathon 2024 for creating an operational management system for Indo-Tibetan Border Police.
- National Finalist at Smart India Hackathon 2025 for developing Fair and Explainable AI Candidate Matching for internships.
- Winner at Silicon Chip Smackdown at (APOGEE 2023, BITS Pilani) for making an AI Poker Bot for No-Limit Texas Hold'em.

TECHNICAL SKILLS

- Programming Languages**: Python, C/C++
- Frameworks and Technologies**: Tensorflow, PyTorch, OpenCV, Transformers, HTML/CSS, FastAPI, LangChain, Git and GitHub
- Cloud Platforms**: Google Cloud Platform, Amazon Web Services, Microsoft Azure